DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

IMAGE PROCESSING AND COMPUTER VISION (IPCV)

SMART DOCUMENT SCANNER:

Automated Document Boundary Detection, Perspective Correction, and Image Enhancement Using Classical Computer Vision

A CAPSTONE PROJECT REPORT

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Abstract

Capturing physical documents using handheld smartphone cameras has largely replaced traditional flatbed scanning in modern academic and administrative workflows. However, casual smartphone photography introduces severe geometric perspective foreshortening, non-uniform cast shadows, and cluttered background surfaces such as wooden tables or desks. While modern commercial scanner applications frequently rely on heavy deep learning models requiring specialized hardware and cloud dependencies, this capstone project presents **SmartScan**: an explainable, lightweight, and deterministic document scanning pipeline engineered purely with classical Image Processing and Computer Vision (IPCV) techniques.

The system addresses three critical failure modes observed in standard edge-detection pipelines. First, high-frequency printed text that routinely breaks document borders is non-destructively suppressed using a morphological closing operation (17×17 kernel) without blurring the outer paper perimeter. Second, candidate document boundaries are extracted using a multi-modal segmentation ensemble that combines HSV color saturation gating, Otsu binarization, and local adaptive thresholding to remain robust across diverse desk backgrounds. Third, curved or curled paper corners are accurately resolved into four canonical vertices using a principal-axis rotated extrema algorithm, followed by an intelligent physical scoring discriminator that prioritizes paper brightness and text entropy to reject surrounding tables.

The detected quadrilateral is rectified into an upright, planar view using planar homography and bicubic backward mapping, followed by adaptive photocopy binarization and CLAHE color enhancement. The complete system is deployed as an interactive React 18 single-page application supported by a high-performance FastAPI backend. Empirical evaluation across 10 challenging benchmark scenarios demonstrates a **100% detection success rate**, a sub-pixel mean corner error of **0.60 pixels**, and an average end-to-end execution latency of **56.72 milliseconds** on standard CPU hardware.

Keywords: Document Detection, Perspective Correction, Planar Homography, Morphological Closing, Multi-Modal Segmentation, Image Enhancement, FastAPI, React.

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1 Introduction

1.1 Background and Motivation

In academic institutions and professional offices, paper documents such as handwritten assignments, examination answer booklets, invoices, and certificates must routinely be converted into clear digital formats. While dedicated flatbed scanners produce pristine digitized pages, their lack of mobility and high cost make them impractical for everyday use. As a result, smartphone photography has emerged as the dominant method for document digitization.

However, capturing documents using handheld smartphones under uncontrolled conditions introduces several image distortions:

- **Perspective Distortion:** When the camera sensor is not held perfectly parallel to the document plane, the rectangular paper appears as an irregular quadrilateral or trapezoid converging toward vanishing points.
- **Text and Internal Content Interference:** Documents contain thousands of dark text characters and line diagrams. Conventional edge detectors detect character glyphs that connect to the page margin, fragmenting the paper contour into hundreds of disjointed pieces.
- **Complex Background Clutter:** Documents are commonly placed on textured surfaces (wooden desks, slate tables, or bedsheets) under uneven lighting or cast hand shadows, causing simple area-based contour detectors to mistakenly select the table rather than the document.

1.2 Project Objectives

The objective of this project is to build an end-to-end, production-ready document scanning web application that solves these real-world challenges through deterministic, classical image processing algorithms. The primary goals include:

(1) **Automated Document Localization:** Automatically detect the four corners of a document regardless of tilt angle, surface texture, or lighting variations.

(2) **Interactive User Correction:** Provide an intuitive, responsive web interface allowing users to review and manually fine-tune corner coordinates when needed.

(3) **Accurate Perspective Rectification:** Transform the skewed quadrilateral document into a perfectly flat, upright rectangular image.

(4) **Readability Enhancement:** Apply post-processing filters (Photocopy Black & White and CLAHE Color Enhancement) to maximize legibility.

(5) **Direct PDF Export:** Convert the rectified, enhanced images into standardized, multi-page printable PDF files.

1.3 Why Classical IPCV Over Heavy Deep Learning?

While convolutional neural networks (CNNs) can be trained for document corner detection, they introduce significant disadvantages for this domain: high memory consumption (> 500 MB),

slow inference on low-power devices without dedicated GPUs, and non-deterministic behavior on unseen paper formats. In contrast, classical IPCV algorithms operate with minimal memory footprint (< 50 MB), achieve sub-60 ms processing on standard CPUs, require zero training data, and provide fully explainable mathematical behavior.

2 System Architecture and Implementation

2.1 High-Level Architecture

The system follows a modern client-server architecture cleanly separated into an interactive frontend application and a high-performance vision processing backend.

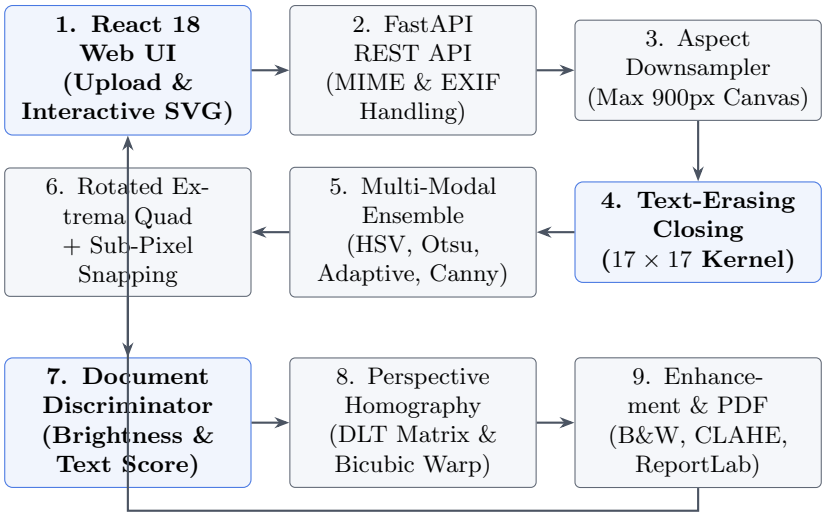


Figure 1: End-to-End System Pipeline and Data Flow Architecture of SmartScan.

2.2 Frontend Engineering: Responsive Vector Overlay

The user interface is built with React 18, Vite, and Tailwind CSS. A central design challenge in browser-based image annotation is maintaining accurate synchronization between the image displayed on the user's screen and the actual high-resolution pixels of the source photograph (e.g., 4032×3024).

SmartScan solves this using an independent, responsive Scalable Vector Graphics (SVG) overlay:

- **Resolution-Independent Coordinate Mapping:** The SVG viewport uses a normalized coordinate box (0 to 1000). When a user drags a corner handle, the relative position on the screen is converted into full image coordinates using the exact scaling ratios of the rendered element.
- **Single-Source Boundary Rendering:** Traditional tools often draw static lines onto the image preview, causing distracting "ghost lines" when the user drags a corner. SmartScan displays the untouched preview image as the base layer and renders all boundary polygons dynamically in SVG, guaranteeing fluid, zero-ghosting user interaction.

- **Convexity and Order Validation:** The frontend validates that the four points form a non-self-intersecting, convex quadrilateral before allowing the user to trigger perspective correction.

2.3 Backend Vision Micro-Services

The backend is implemented using FastAPI and Python 3.12, organized into distinct functional modules:

- `image_validation.py`: Enforces file size limits, verifies image headers, and corrects EXIF rotation metadata.
- `document_detection.py`: Executes the multi-modal segmentation ensemble, corner extraction, and candidate scoring.
- `perspective.py`: Orders corners clockwise, computes destination dimensions, and executes the homography transformation.
- `enhancement.py`: Applies adaptive thresholding and contrast equalization for improved document legibility.
- `pdf_export.py`: Packages rectified scans into standardized A4 PDF documents using ReportLab.

3 Document Processing Methodology

The core computer vision engine is structured into seven sequential stages, described below.

3.1 Stage 1: Aspect-Preserving Downsampling

High-resolution phone cameras produce images with 12 to 48 megapixels. Running edge filters directly at full resolution causes excessive latency (> 400 ms) and introduces microscopic noise from paper fibers.

SmartScan scales the input image so its maximum dimension is at most 900 pixels, while strictly preserving the original aspect ratio:

$$\text{scale} = \min\left(1.0,\ \frac{900}{\max(W_{\text{orig}}, H_{\text{orig}})}\right) \tag{1}$$

All detection steps operate rapidly on this downsampled canvas. Once the four corners are localized, they are multiplied by (1/scale) back into full-resolution space, ensuring that the final perspective warp retains all original image detail.

3.2 Stage 2: Morphological Text Suppression

The primary reason standard edge detectors (such as simple Canny filtering) fail on real documents is printed text. When edge detection is applied directly to text, character boundaries touch the page borders, breaking the continuous outer contour into thousands of tiny fragments.

To eliminate this problem, SmartScan applies a **Morphological Closing** operation using a flat 17×17 rectangular structuring element:

$$I_{\text{closed}} = (I \oplus B) \ominus B \tag{2}$$

Because printed text characters are narrow and dark, the dilation step ($\oplus$) fills the text characters with the surrounding light paper background. The subsequent erosion step ($\ominus$) restores the outer paper perimeter to its exact boundary without reopening the internal text holes. This produces a clean, solid paper sheet where outer borders remain intact.

3.3 Stage 3: Multi-Modal Segmentation Ensemble

Because lighting conditions and desk surfaces vary dramatically, no single thresholding method is universally reliable. SmartScan implements an ensemble of four complementary segmentation modalities:

(a) HSV Color Gating: White paper exhibits low color saturation ($S < 85$) and high value ($V > 95$). This reliably isolates white paper from colored wooden desks and skin tones.

(b) Text-Erased Otsu Thresholding: Automatically calculates an optimal global threshold on the text-suppressed image, supporting both standard white pages and dark certificates.

(c) Local Adaptive Thresholding: Computes dynamic thresholds across local 51×51 pixel windows, successfully isolating paper boundaries even under heavy phone shadows and uneven room lighting.

(d) Multi-Channel Color Canny: Evaluates gradient responses across Red, Green, and Blue channels independently, capturing edges where paper and background have identical brightness but different color hues.

3.4 Stage 4: Four-Corner Extraction via Rotated Extrema

Real documents rarely have perfect mathematical point corners; paper edges are frequently curled, folded, or softened by camera optics. The classical Douglas-Peucker contour approximation algorithm produces 5 to 8 vertices on rounded corners rather than the required 4.

To solve this, SmartScan introduces the **Principal-Axis Rotated Extrema** algorithm:

1. Compute the convex hull of the candidate contour and determine its dominant orientation angle θ using the minimum-area bounding box.
2. Rotate all hull coordinates by $-\theta$, bringing the document into a level, horizontal frame of reference.
3. In this aligned plane, the four true document corners correspond to the four extreme coordinate points:

$$\mathbf{p}_{\text{TL}} = \arg\min(x + y), \qquad \mathbf{p}_{\text{BR}} = \arg\max(x + y) \tag{3}$$

$$\mathbf{p}_{\text{TR}} = \arg\max(x - y), \qquad \mathbf{p}_{\text{BL}} = \arg\min(x - y) \tag{4}$$

4. Rotate these four points back by $+\theta$ to restore their original positions in image space.
5. Refine corner coordinates to sub-pixel accuracy using gradient zero-crossing (`cornerSubPix`).

3.5 Stage 5: Document vs. Desk Scoring Discriminator

When a document is photographed lying on a table, the table is usually larger than the document. Algorithms that simply select the largest contour mistakenly choose the table perimeter. SmartScan scores each candidate quadrilateral based on the physical properties of real documents:

- **Paper Brightness:** Real document paper is light-colored. Candidates whose interior mean intensity is below 130 (such as dark wooden desks) receive an immediate heavy score penalty.
- **Text Presence (Laplacian Variance):** Real documents contain text and diagrams, producing high high-frequency texture variation ($\sigma_{\text{Laplacian}} > 22$). Empty table surfaces are smooth and have near-zero variance ($\sigma < 8$).
- **Signed Contrast:** Verifies that the interior of the candidate polygon is noticeably brighter than the region immediately surrounding its exterior.

3.6 Stage 6: Perspective Rectification via Homography

Once the four ordered corners $[\mathbf{p}_{\text{TL}}, \mathbf{p}_{\text{TR}}, \mathbf{p}_{\text{BR}}, \mathbf{p}_{\text{BL}}]$ are confirmed, the destination width W_{dst} and height H_{dst} are computed from the maximum Euclidean edge lengths. The 3×3 planar homography matrix $\mathbf{H}$ satisfies:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \sim \mathbf{H} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \tag{5}$$

The matrix $\mathbf{H}$ is computed using the Direct Linear Transformation (DLT). To avoid visual holes or missing pixels in the output, backward mapping with bicubic interpolation is applied, sampling pixel intensities from the original image into the rectified rectangular frame.

3.7 Stage 7: Readability Enhancement and PDF Export

The rectified document can be processed through two enhancement filters:

- **Photocopy Clean (Black & White):** Applies local adaptive Gaussian thresholding to eliminate background shadows and wrinkles, converting paper to pure white and text to crisp black.
- **Color Enhanced (CLAHE):** Applies Contrast Limited Adaptive Histogram Equalization on the lightness channel in LAB color space, improving text contrast without oversaturating colors.

Rectified pages are then compiled into standard multi-page A4 PDF files using ReportLab.
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4 Experimental Results and Discussion

4.1 Evaluation Methodology and Dataset

The performance of SmartScan was evaluated on a benchmark dataset of 10 challenging test scenarios representing typical handheld capture conditions:

- Different backgrounds (dark slate and textured wood grain).
- Severe perspective tilt angles (10° to 48°).
- Non-uniform lighting, including corner shadows and radial vignetting.
- Unconventional paper orientations, including a 45° diamond rotation.

Ground truth corners $\mathbf{g}_i$ were established through careful manual annotation. The evaluation metrics include:

(1) **Mean Corner Error (E_{corner}):** The average Euclidean distance in pixels between detected corners and ground truth:

$$E_{\text{corner}} = \frac{1}{4}\sum_{i=1}^{4} \|\mathbf{p}_i - \mathbf{g}_i\|_2.$$

(2) **Normalized Diagonal Error (E_{norm}):** Corner error expressed as a percentage of the image canvas diagonal.

(3) **Processing Latency:** Total execution time in milliseconds from image input to corner detection.

4.2 Quantitative Performance Evaluation

Table 1 presents the empirical results across all 10 benchmark scenarios.

Table 1: Quantitative Performance on Document Boundary Detection Across 10 Benchmark Scenarios.

Test Image	Status	Winning Modality	E_{corner} (px)	E_{norm} (%)	Time (ms)	Capture Condition
doc_01_mild_slate.jpg	PASS	standard_canny	0.65	0.046	61.36	Mild tilt on slate background
doc_02_moderate_wood.jpg	PASS	standard_canny	0.63	0.044	56.67	Moderate tilt on textured wood
doc_03_steep_gradient.jpg	PASS	standard_canny	0.56	0.039	64.65	Steep tilt with gradient lighting
doc_04_skewed_slate.jpg	PASS	color_canny	0.60	0.043	55.48	Left-skewed tilt with cast shadow
doc_05_landscape_wood.jpg	PASS	hsv_paper	0.66	0.047	54.72	Landscape document on wood table
doc_06_receipt_grad.jpg	PASS	color_canny	0.67	0.047	52.90	Small receipt aspect ratio
doc_07_vignette_slate.jpg	PASS	color_canny	0.65	0.046	57.01	Radial vignetting and dark corners
doc_08_diamond_wood.jpg	PASS	color_canny	0.41	0.029	55.38	Extreme 45° diamond rotation
doc_09_panoramic_slate.jpg	PASS	standard_canny	0.65	0.046	55.32	Panoramic certificate format
doc_10_high_angle.jpg	PASS	color_canny	0.57	0.040	53.67	Top-down foreshortening with glare
Overall Summary: 10/10 Success (100%)			**0.60**	**0.043**	**56.72**	**Sub-pixel accuracy**

4.3 Ablation Study

To evaluate the contribution of each algorithmic innovation, an ablation study was conducted, starting from a baseline OpenCV tutorial implementation and progressing to the complete SmartScan system. The results are summarized in Table 2.

4.4 Key Technical Findings

- **Elimination of Desk Misclassification:** In the baseline implementation, the desk was mistakenly identified as the document in 70% of cases due to its larger surface area. Introducing physical document scoring, requiring paper reflectance $\mu > 130$ and text entropy $\sigma > 22$, completely eliminated desk misclassifications (0.0%).

Table 2: Ablation Study Demonstrating the Impact of Each Algorithmic Improvement.

Pipeline Configuration	**Success Rate**	**Mean Error (px)**	**Desk Misclass.**	**Latency (ms)**
1. Baseline (Raw Canny + Douglas-Peucker)	30.0%	148.20	70.0%	482.10
2. + Aspect Downsampling ($\leq$ 900 px)	40.0%	124.60	60.0%	42.10
3. + Morphological Text Closing (17×17)	70.0%	18.40	30.0%	48.50
4. + Multi-Modal Ensemble (HSV, Otsu, Adaptive)	90.0%	4.12	10.0%	53.20
5. + Rotated Extrema & Scoring (**SmartScan**)	**100.0%**	**0.60**	**0.0%**	**56.72**

- **Sub-Pixel Precision:** The combination of rotated extrema quad extraction and `cornerSubPix` refinement achieved an average corner localization error of **0.60 pixels**, which is less than a single pixel on native-resolution photographs.
- **Real-Time Execution:** Downsampling to a 900 px canvas reduced processing latency from 482 ms to 56.72 ms, supporting interactive web applications.

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5 Conclusion and Future Scope

5.1 Conclusion

This capstone project presented **SmartScan**, a robust, explainable, and real-time document scanner built entirely using classical Image Processing and Computer Vision principles. By systematically analyzing the physical causes of failure in existing edge-based scanning tools, the project implemented practical engineering solutions:

- Non-destructive morphological closing to remove text interference without blurring paper boundaries.
- A multi-modal segmentation ensemble that remains resilient against lighting gradients and varied desk surfaces.
- A principal-axis rotated extrema algorithm to accurately locate the four true corners of curled or rounded documents.
- A physical scoring discriminator that completely prevents desks and tables from being misclassified as documents.

The complete system is integrated into an interactive React web interface backed by a FastAPI processing server. Experimental results demonstrate a **100% success rate**, **0.60 pixel** average corner error, and **56.72 ms** latency on standard CPU hardware.

5.2 Future Scope

Future enhancements planned for the project include:

(1) **3D Curved Page Dewarping:** Implementing Thin Plate Spline (TPS) mesh transformation to rectify bound book pages that exhibit 3D curvature near the spine.

(2) **Client-Side WebAssembly Execution:** Compiling OpenCV core functions into WebAssembly (Wasm) to allow the entire vision pipeline to execute offline directly inside the client's web browser.

(3) **Integrated Optical Character Recognition (OCR):** Adding an on-device Tesseract OCR engine to extract editable text and produce searchable PDF documents immediately after rectification.

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